

Galaxy-X-os — One-Page Technical Report

SCALE x ODYSSEY | Deep-learning classification of raw astronomical images into 5 celestial categories.

GitHub: <https://github.com/Srujan0798/Galaxy-X-os>

Dataset Sources

- **Spiral Galaxy + Elliptical Galaxy** — real survey imagery from **Galaxy10 DECaLS** (via astroNN; real SDSS/DECaLS images), mapped from Galaxy10's 10 morphology classes into our two galaxy-morphology classes.
- **Nebula + Star Cluster + Planetary Object** — real imagery from the **NASA Image Library** (images.nasa.gov, no API key): Hubble/Spitzer/JPL mission images retrieved by keyword with per-class purity filters. Implemented in `src/download_archives.py`. Kaggle sets + a labelled procedural fallback remain as fallbacks only.
- **5 classes:** Spiral Galaxy, Elliptical Galaxy, Nebula, Star Cluster, Planetary Object.
- **Split:** stratified disjoint train/val/test, verified by MD5 hash that no image appears in more than one split (zero leakage).
- **Source of truth:** `data/processed/DATA_MANIFEST.json` — all five classes real: Galaxy10 DECaLS for galaxies, NASA Image Library for the other three.

Model Architecture

- Backbone: EfficientNet-B3 (ImageNet pretrained, transfer learning) — ~11.6M parameters.
- Head: replaced classifier → 5-class output.
- Training: progressive unfreezing (backbone frozen 3 epochs), mixed-precision (torch.amp), AdamW (lr 3e-4), batch 32, OneCycleLR.
- Augmentations: Albumentations astro-specific pipeline (cosmic-ray sim, vignetting, Poisson noise, label smoothing 0.1).
- Explainability: Grad-CAM heatmaps.

Final Test Metrics

Evaluated on the held-out disjoint test set (249 images, ~50 per class — none seen in training). Macro-averaged. **All five classes trained on real telescope imagery.**

Metric	Standard	+ TTA (6× aug)
Accuracy	93.17%	92.77%
Precision (macro)	0.934	—
Recall (macro)	0.932	—
F1 (macro)	0.932	0.928

Per-class F1 (Standard):

Class	F1
Spiral Galaxy	0.887
Elliptical Galaxy	0.895
Nebula	0.949
Star Cluster	0.947
Planetary Object	0.980

Spiral vs Elliptical show the realistic ~0.88–0.90 F1 of genuine morphological ambiguity. Nebula / Star Cluster / Planetary, now on real NASA imagery, show realistic 0.95–0.98 (the earlier perfect 1.000 reflected clean procedural data).

Confusion Matrix

See `results/confusion_matrix.png`. Additional plots: `results/per_class_metrics.png`, `results/confidence_distribution.png`.

Inference Time

Measured ~72 ms per image (median; 118 ms mean) on Apple MPS (EfficientNet-B3, batch=1, after warmup) — well under the 5 s requirement. Faster on CUDA GPU. Streamlit demo runs in real time on consumer hardware.

Setup Instructions

```
pip install -r requirements.txt
python src/prepare_data.py # downloads real data (no API key)
python src/train.py
python src/evaluate.py # standard + TTA
python src/gradcam.py
streamlit run app/app.py
```

One-click alternative: open notebooks/Galaxy_X_Colab.ipynb in Google Colab to run the full pipeline end-to-end on a free GPU.

Training Script

Full pipeline: src/train.py — progressive unfreezing + OneCycleLR, AdamW (lr 3e-4), batch 32, mixed precision (AMP), class weights, label smoothing. Config: configs/config.yaml. Checkpoint: checkpoints/best_model.pth.

Training note: the reported model was trained to 93.17% / 92.77% on a CUDA GPU (Google Colab, T4) over a 50-epoch schedule via notebooks/Galaxy_X_Colab.ipynb on fully-real imagery (Galaxy10 DECaLS + NASA Image Library). Inference runs on consumer hardware — ~72 ms/image on an Apple-MPS MacBook.